



1st Semester Exam, 2024/2025

Exercise 1

(06 marks)

Consider the following program:

```
1 #include <stdio.h>
2 int main(){
3     int found = 0;
4     const int N=5;
5     int X;
6     int T[]={5,2,4,8,9};
7     int index = -1;
8     scanf("%d", ...x);
9     for (... i = 0; i ... N; i++) {
10         if (T[i] == x) {
11             found = 1;
12             ... =i;
13             break;
14         }
15     }
16     printf("the result is %d at index
17           %d",found, index);
18     return 0;}
```

- 1 ☐ Fill in the gaps in this code(01.5 marks)
- 2 ☐ There is one error in this code, Cite and correct it.(0.5 mark)
- 3 ☐ Give the trace of this program for x=8.(0.5 marks)
- 4 ☐ What does this code do?(0.5 mark)
- 5 ☐ When we replace the line 10 by: `if (T[i] = x){`, with x=3, the program will:(01 mark)
 - (a) not compile and displays an error.
 - (b) compile but does not display the correct result .
 - (c) compile and always displays the correct result.
 - (d) modify the array elements.
- 6 ☐ Rewrite the code using the while loop(01 mark)
- 7 ☐ Rewrite the code without using the keyword break(01 mark)

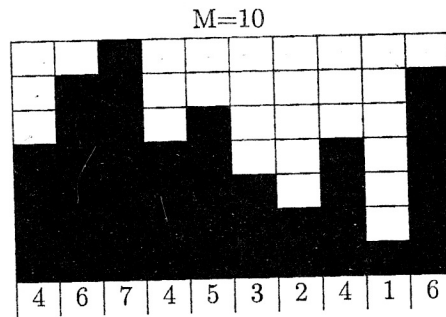
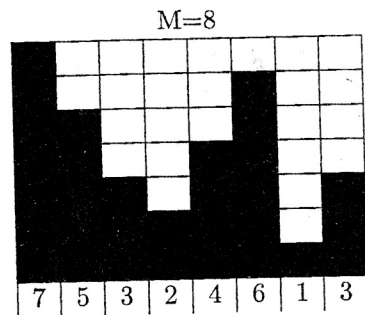
Exercise 2

(08 marks)

Given W, a set of M vertical bars, each bar has a non-negative integer size, consider the following examples:

Write a C program to display :

- ☐ black_area, the area of the black zone in W.(01.5 marks)
- ☐ white_area, the area of the white zone in W.(01.5 marks)
- ☐ num_descends, how many times you go down when crossing the bars in W.(02 marks)
- ☐ large_rect, the area of the largest rectangle in W (All squares are 1x1).(03 marks)



Exercise 3 (06 marks)

Given an array L of p integers.

Write a program to find the minimum number of changes in the array so that the array will be strictly in increasing order.

Examples:

Input :

0	1	8	3	4
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Output: 2

We can change $L[2]$ to any value between 2 and 8 and $L[4]$ to any value greater than 4.

Input :

1	2	3	5	7	9
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Output : 0

L is already strictly increasing.

N.B. Avoid creating solutions that only work for the provided examples.